

Polybrominated diphenyl ethers (PBDEs), hydroxylated PBDEs (OH-PBDEs), and measures of thyroid function in second trimester pregnant women in California



Prenatal exposure to polybrominated diphenyl ethers (PBDEs) may disrupt thyroid function and contribute to adverse neurodevelopmental outcomes. We conducted a pilot study to explore the relationship between serum concentrations of lower-brominated PBDEs (BDE-17 to -154), higher-brominated PBDEs (BDE-183 to -209), and hydroxylated PBDE metabolites (OH-PBDEs) with measures of thyroid function in pregnant women. Concentrations of PBDEs, OH-PBDEs, thyroid-stimulating hormone (TSH), total thyroxine (T4), and free T4 were measured in serum samples collected between 2008 and 2009 from 25 second trimester pregnant women in California. Median concentrations of lower-brominated PBDEs and OH-PBDEs were the highest reported to date in pregnant women. Median concentrations of BDE-47 and the sum of lower-brominated PBDEs (Σ PBDE5) were 43.1 ng/g lipid and 85.8 ng/g lipid; and 0.084 ng/mL for the sum of OH-PBDEs (Σ OH-PBDE4). We observed a positive association between the weighted sum of chemicals known to bind to transthyretin (Σ TTR binders) and TSH levels. We also found positive associations between TSH and Σ PBDE5, Σ OH-PBDE4, BDE-47, BDE-85, 5-OH-BDE47, and 4'-OH-BDE49; and an inverse association with BDE-207. Relationships with free and total T4 were weak and inconsistent. Our results indicate that PBDE exposures are elevated in pregnant women in California, and suggest a relationship with thyroid function. Further investigation is warranted to characterize the risks of PBDE exposures during pregnancy.